PRODEMAND

YMMS:	2008 Mazda 3 s Sport		May 20, 2024
Engine:	2.3L Eng	License:	
VIN:	JM1BK343781150072	Odometer:	

Input Speed Sensor,Transmission

Operation

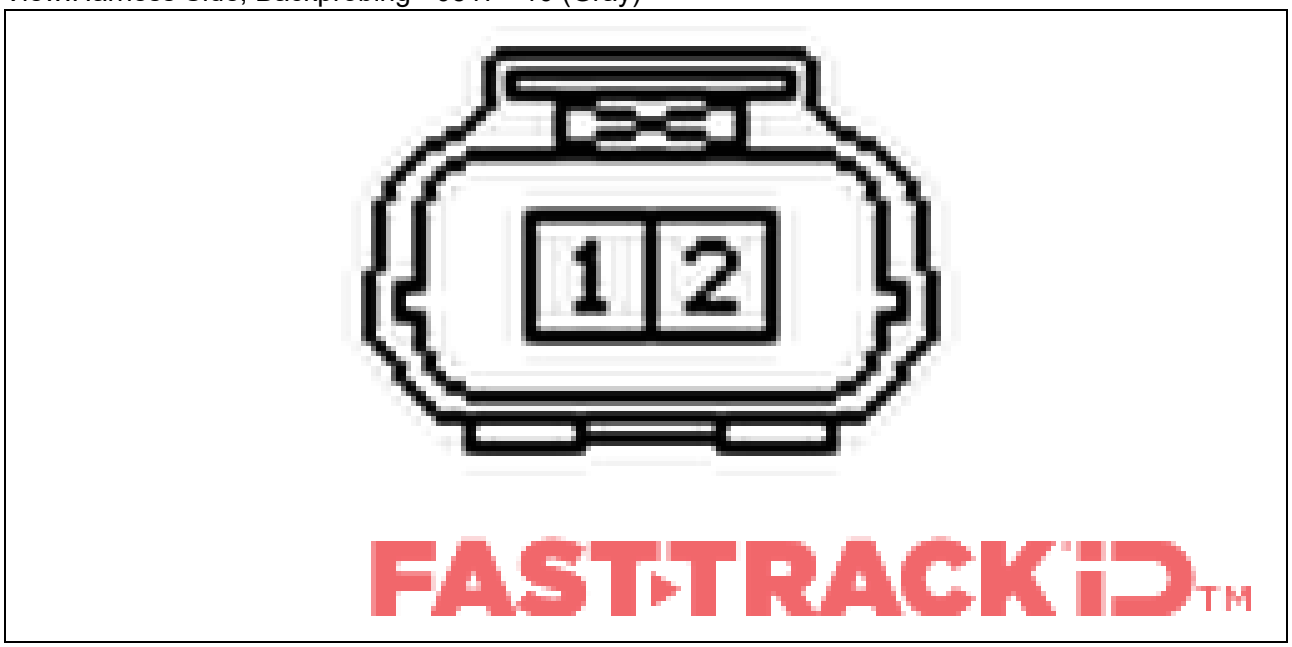
This sensor is an Hall - Effect style sensor that measures the rotational speed of the input shaft. When the tooth of the reluctor wheel passes through the magnetic field created by the sensor, voltage drops low. When the tooth passes and the sensor measures a gap between the reluctor teeth, voltage rises. As engine RPM increases, the frequency of this signal will increase.
Input/turbine speed sensor reads reverse clutch drum & produces an AC voltage signal (sine Wave). As Engine RPM increases, frequency & amplitude increase.
TCM uses input/turbine speed sensor, intermediate sensor and vehicle speedometer sensor signals to monitor trans gear ratio.
Erratic input/turbine speed sensor operation may cause shifting, line pressure and TCC lock up concerns.

Location

Best Test Location:
2 pin connector on input/turbine speed sensor.
Component Location:
Top of transmission under battery.

Connector

View:Harness Side, Backprobing - 0517 - 10 (Gray)



Pin Assignment	Wire Color

1=Input Speed Sensor Signal	Yellow
2=Input Speed Sensor Ground	Brown

Tests

Frequency Test

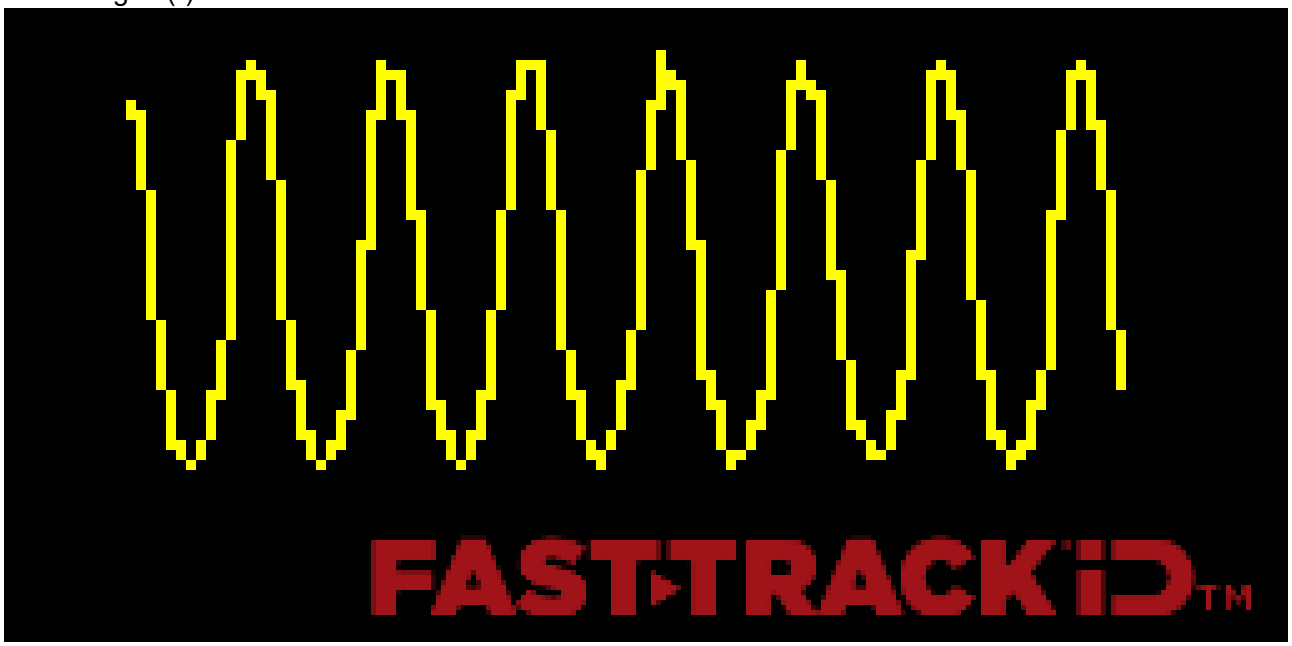
Back probe connector
Yellow=Signal(+)
Black=Signal (-)
Monitor pulse generator signal while driving.
Frequency is sine Wave and varies with Engine speed.

Resistance Test

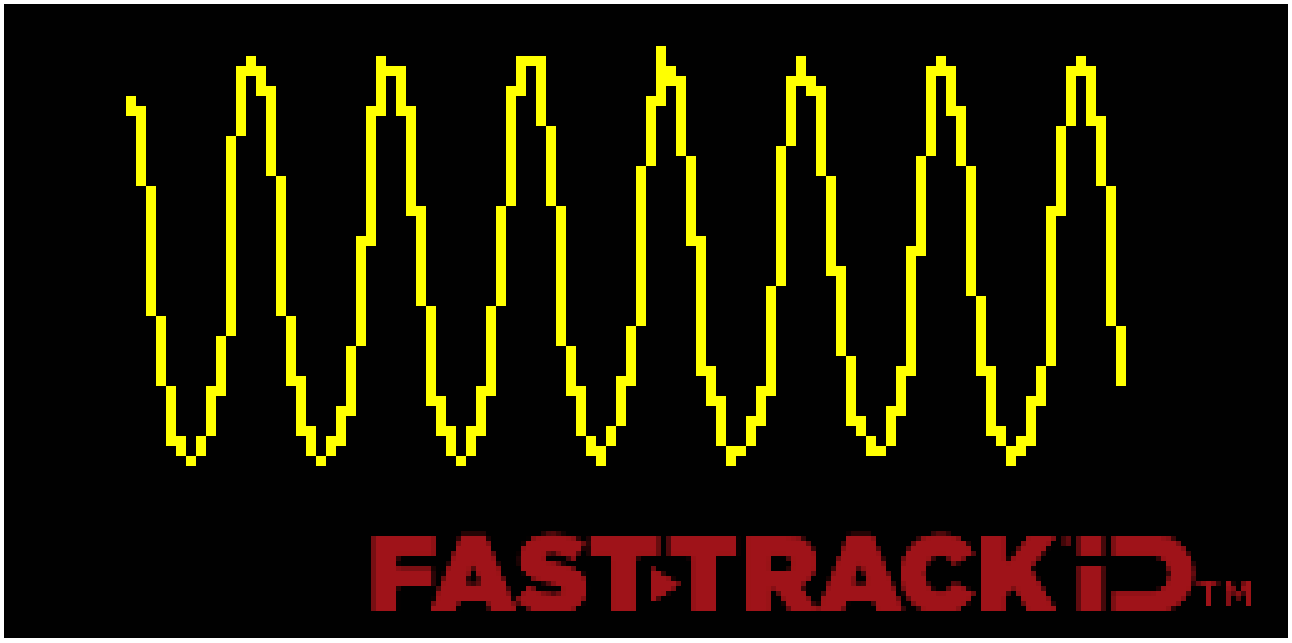
Key off, disconnect sensor.
Connect test leads across sensor.
Resistance should be about:
250Ω to 600Ω.

Signature Test

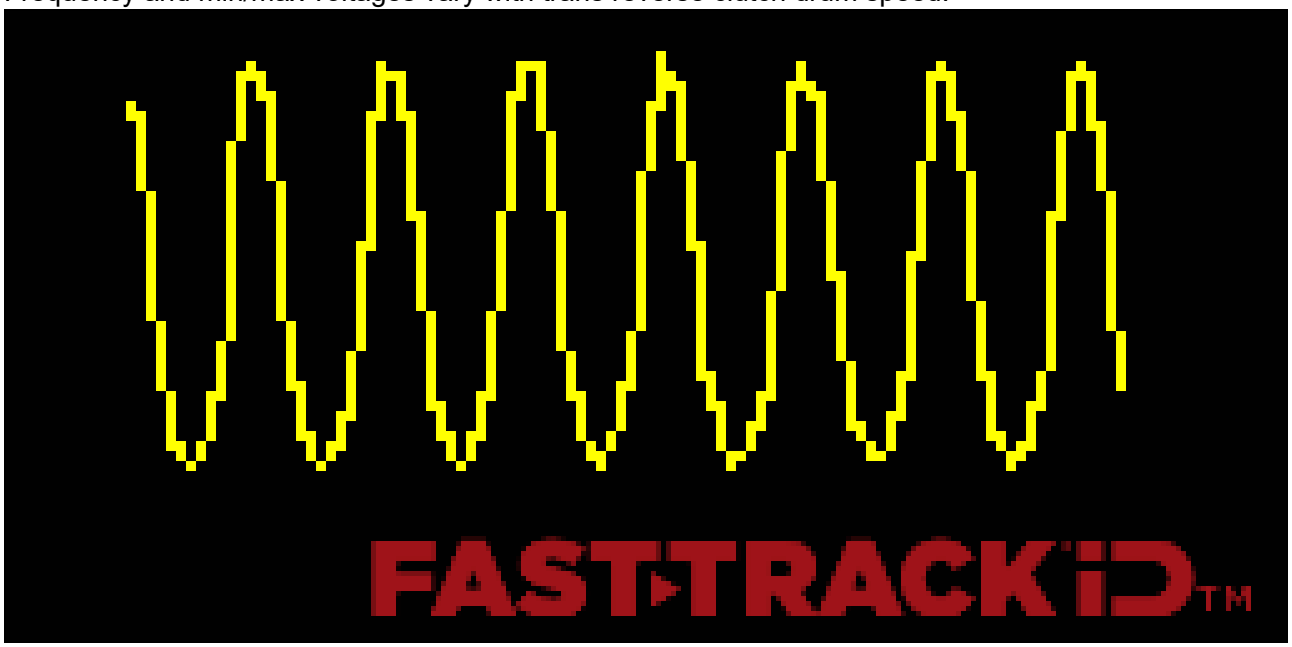
Connect test leads:
Yellow=Signal(+)
Black=Signal(-)



Use Lab Scope to check waveform signature integrity.



Frequency and min/max voltages vary with trans reverse clutch drum speed.



Intermediate Speed Sensor,Transmission

Operation

Intermediate sensor reads output gear teeth & produces a digital (on/off) voltage signal.
TCM uses intermediate sensor, vehicle speedometer sensor, and input/turbine speed sensor signals to monitor trans gear ratio.
Erratic intermediate sensor operation may cause shifting concerns.

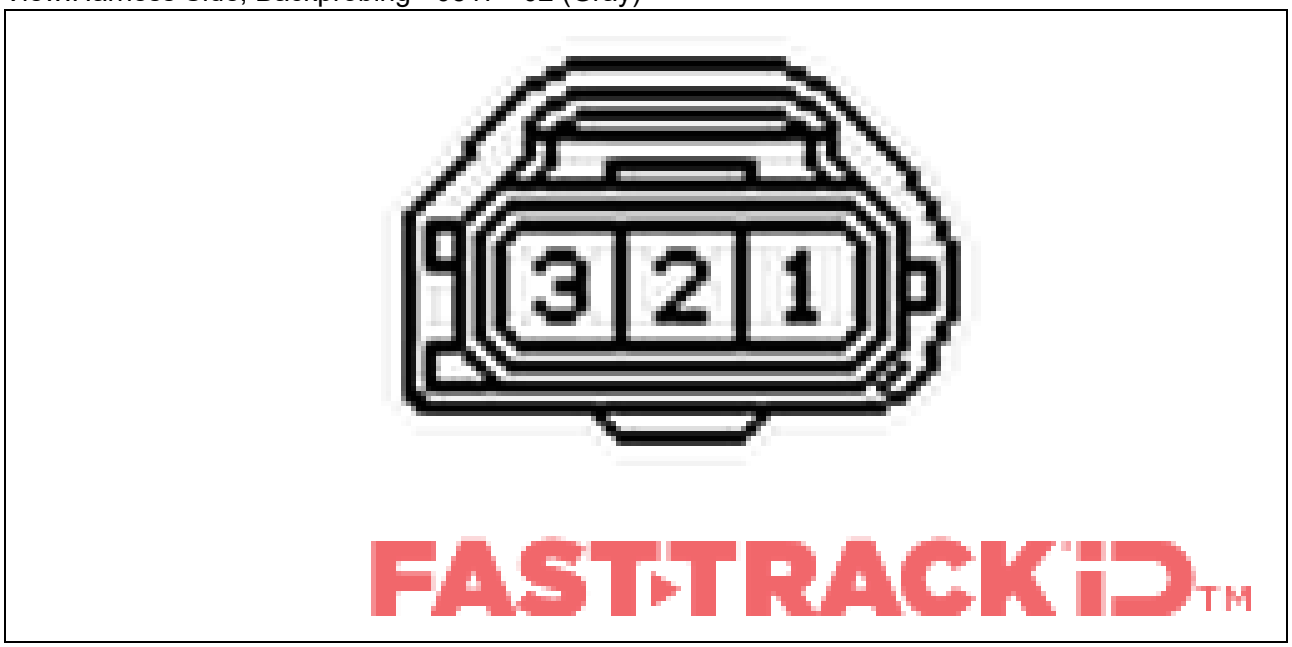
Location

Best Test Location:
Grey 3 pin connector at sesor of trans toward firwall.
Component Location:

Top of trans toward firwall.

Connector

View:Harness Side, Backprobing - 0517 - 02 (Gray)



Pin Assignment	Wire Color
1=Ignition Power	Yellow
2=Intermediate Sensor Signal	Orange
3=Ground	Black

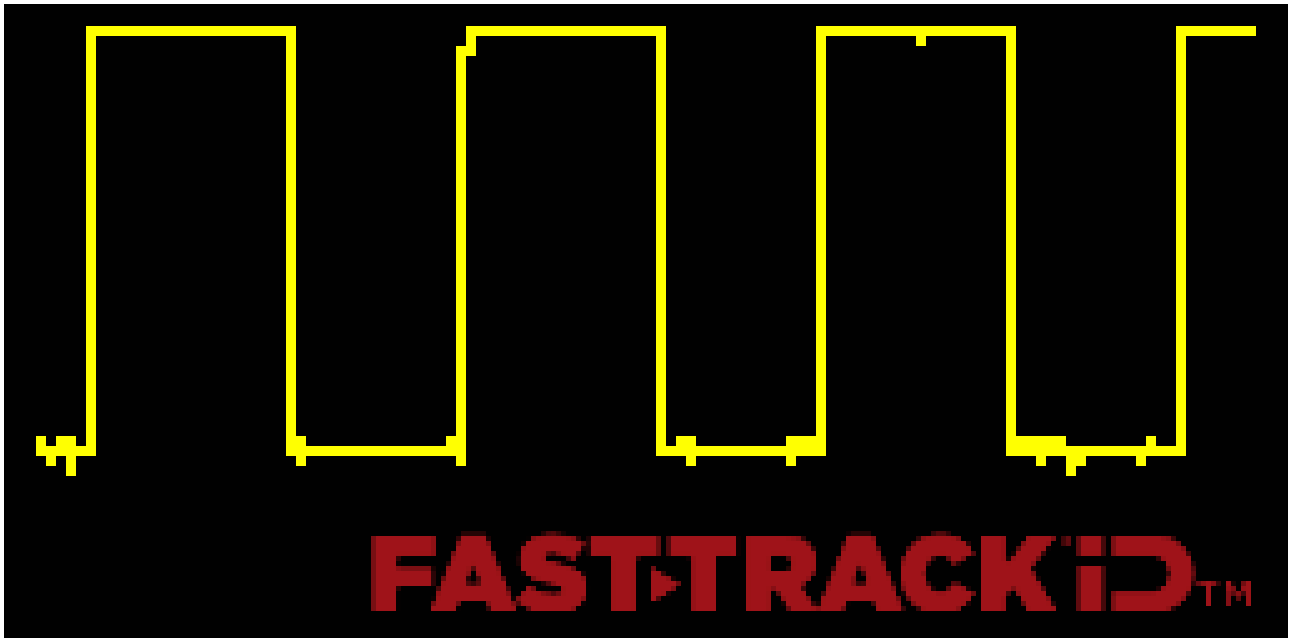
Tests

Frequency Test

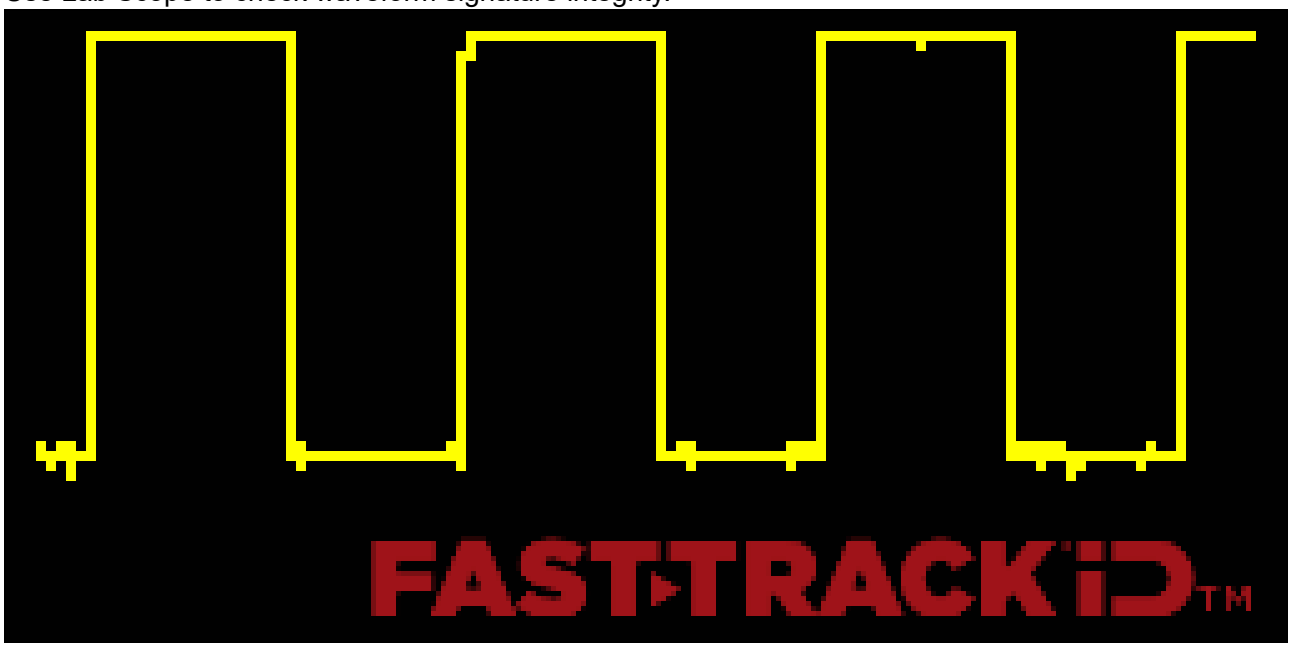
Backprobe connector
Yellow=intermediate sensor signal
Black=known good ground
Frequency should show smooth rise with RPM increase and decrease with no drop outs or glitches.
Frequency should change when trans shifts gears.

Signature Test

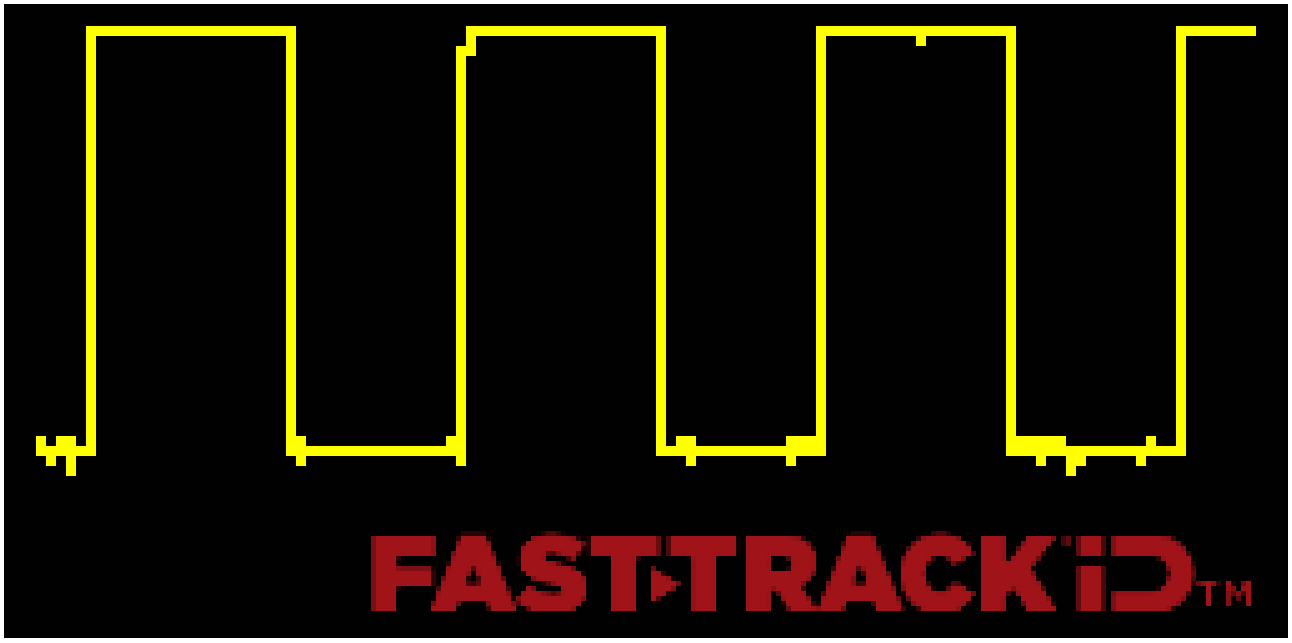
Connect test leads:
Yellow=intermediate sensor signal
Black=known good ground



Use Lab Scope to check waveform signature integrity.



Frequency varies with trans speed.



Wheel Speed Sensors,ABS

Operation

This sensor is a Hall-Effect type switch that measures vehicle speed by measuring a reluctor wheel located at each wheel. As the wheel spins, a digital square wave signal is produced. There is no separate signal wire; the signal is superimposed onto both signal wires. This signal frequency will increase with an increase in wheel speed.
The ABS module uses this signal as a vehicle speed input and, by monitoring the WSS signals, can detect wheel slippage. This signal may also be used for cruise control.
ABS WSS use a semiconductor Element with an active drive circuit that changes the AC signal to a square Wave.
WSS sends signal to ABS Control module. ABS Control module then sends VSS signal to PCM & instrument cluster via can network.
Do not use ohm meter to check WSS resistance or damage to the semiconductor could occur.

Location

Best Test Location:
2 pin connector at the wheel speed sensor.
Component Location:
Front: top of spindle
Rear: inboard side of hub.

Connector

View: Harness Side, Backprobing - (Black)



Pin Assignment	Wire Color
1=WSS Signal	White
2=WSS Ground	Gray

Tech Notes

CAUTION, Do not use ohm meter to check WSS resistance or damage to the semiconductor could occur.

Tests

Frequency Test

Connect test leads:
Yellow=WSS signal
Black=known good ground
Monitoring WSS signal while driving.
As vehicle speed increases WSS frequency will increase.
Frequency should change smoothly with no dropouts or glitches during test.

Signature Test

Connect test leads:
Yellow=WSS signal
Black=Known good ground
Connector will be displayed here.
Lab Scope should be used to check waveform signature integrity



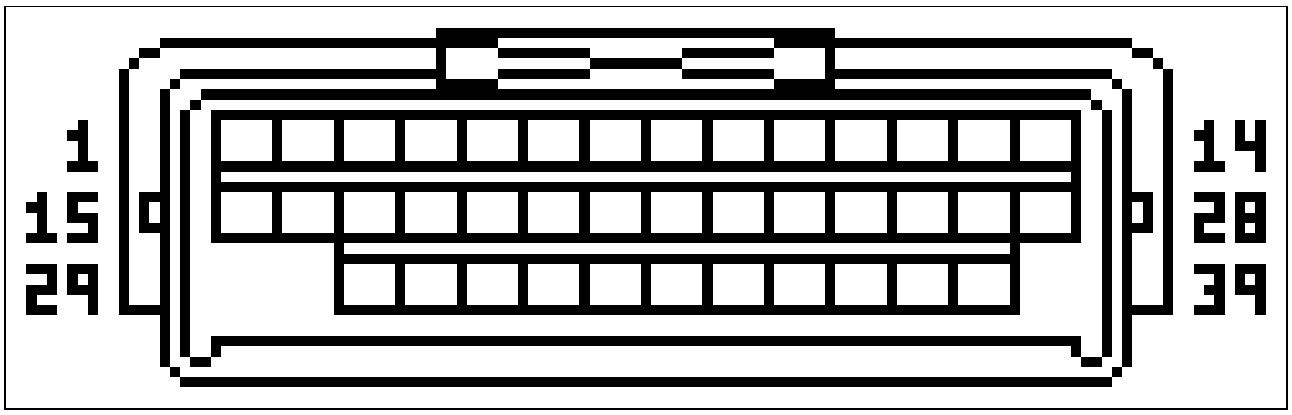
Rotate the tire by hand, as the speed of the spinning wheel changes the waveform frequency will change.



TCM,Transmission

Pinouts

View:Backprobing Harness Connector, PCM Connector - 0517 - 01 (Gray)



Pin Assignment	Wire Color
2=CAN Module	White
5=CAN Module	Red
6=Down Shift	Violet
7=Up Shift	White
9=Main Relay	Yellow
10=Battery Power	Green
11=M Position Switch	Orange
12=Ground	Black
13=Ground	Black
15=Main Relay	Yellow
16=Main Relay	Blue
19=Oil Pressure Switch	Orange
21=TR Switch	Gray
22=TR Switch,TFT Sensor Ground	Pink
25=Input/Turbine Speed Sensor Ground	Brown
26=VSS	Red
27=TFT Sensor Signal	LtGreen
28=Input/Turbine Speed Sensor Signal	Yellow
29=Intermediate Sensor	Orange
30=Line Pressure Control Solenoid A Signal	Orange
31=Line Pressure Control Solenoid A Ground	Blue
33=Shift Solenoid A	SkyBlue
34=Shift Solenoid D	Brown

35=Shift Solenoid F	Pink
36=Shift Solenoid B	Yellow
37=Shift Solenoid E	White
38=Shift Solenoid C	Gray
39=Line Pressure Control Solenoid B	Blue